Glyph Mini 2040 MIDI Keyset

Oct 22, 2026 · @Dinesh v

Overview

The keyset is a USB MIDI controller built on a Pcbcupid Glyph Mini 2040 (RP2040). It plugs into a laptop or phone, appears as the MIDI device **Glyph Mini 2040**, and needs no drivers.

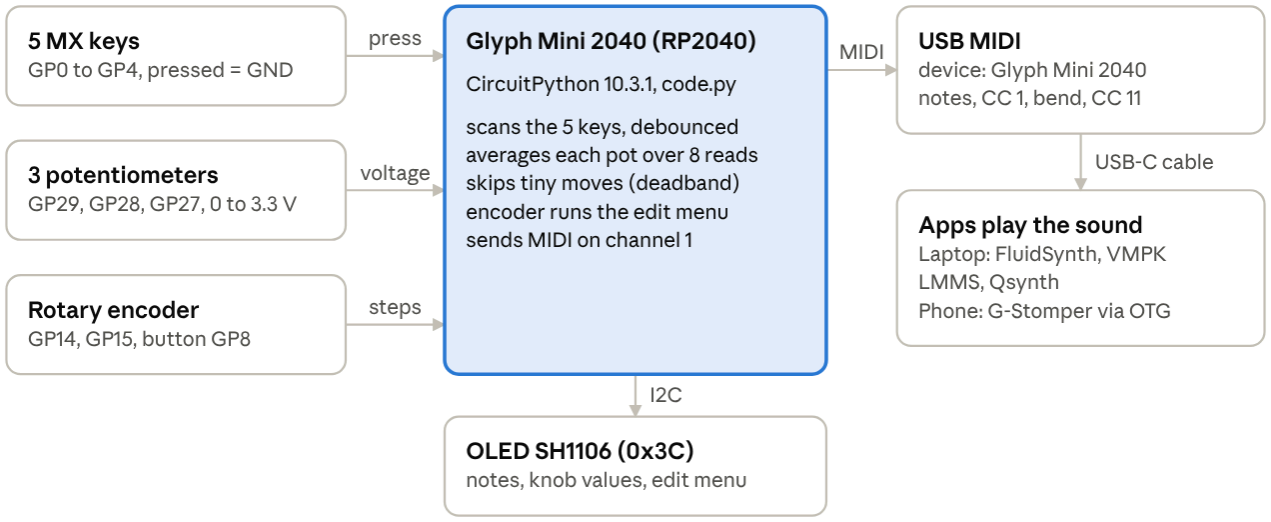
- **5 MX key switches** play notes.
- **3 knobs (potentiometers)** send modulation, pitch bend and expression.
- **A rotary encoder with push button** drives an on-screen menu to change each key's note, velocity and channel, and each knob's settings.
- **A 1.3″ SH1106 OLED** shows the knob values, the key notes and the edit menu.

The firmware is Adafruit's MIDI Keyset code (CircuitPython), adapted to this board's pins. All parts were tested on the board on 22 Oct 2026.

How it works

The Glyph reads every control, turns it into a MIDI message in `code.py`, and sends it over the USB cable; the app on the other end makes the sound.

Every control becomes a USB MIDI message inside the Glyph



signal flow · 3 inputs, the RP2040, USB MIDI out

The keyset itself makes no sound: without an app such as FluidSynth or G-Stomper listening, the messages go nowhere. The OLED only shows state; it is not needed for MIDI to work.

Parts list

Part	Qty	Notes
Pcbcupid Glyph Mini 2040	1	RP2040, USB-C, runs CircuitPython 10.3.1 (Glyph Mini build)
MX key switch	5	any MX-style switch; 2 pins, no polarity
Rotary encoder with push switch	1	bare 5-pin type: 3 pins for rotation, 2 for the button
Potentiometer, 10 kΩ linear (B10K)	3	3 legs: outer, wiper (middle), outer
OLED 1.3″ 128×64, SH1106, I2C	1	4 pins: GND, VCC, SCL, SDA; address 0x3C
Protoboard	1	perfboard with the Glyph on headers
USB-C data cable	1	plus a USB-C OTG adapter for a phone

No resistors, diodes or extra parts are needed: the RP2040's internal pull-ups handle the keys, encoder and button.

Pinout

The keyset uses 14 of the Glyph Mini 2040's pins; GP5, GP9–GP13 and GP26 are still free.

Pin	Used for	Notes
GP0	Key 1	to GND when pressed
GP1	Key 2	to GND when pressed
GP2	Key 3	to GND when pressed
GP3	Key 4	to GND when pressed
GP4	Key 5	to GND when pressed
GP6	OLED SDA	I2C1 data; must pair with GP7
GP7	OLED SCL	I2C1 clock

Pin	Used for	Notes
GP8	Encoder button	the board's labelled SDA pin, used here as a plain input
GP14	Encoder A	must sit next to GP15
GP15	Encoder B	swap with GP14 to reverse direction
GP27	Pot C wiper	analog input (ADC1)
GP28	Pot B wiper	analog input (ADC2)
GP29	Pot A wiper	analog input (ADC3)
3V3	Power rail	OLED VCC and one outer leg of each pot
GND	Ground rail	keys, encoder, button, OLED, pots
GP5, GP9, GP10, GP11, GP12, GP13, GP26	Free	GP26 is the only free analog pin

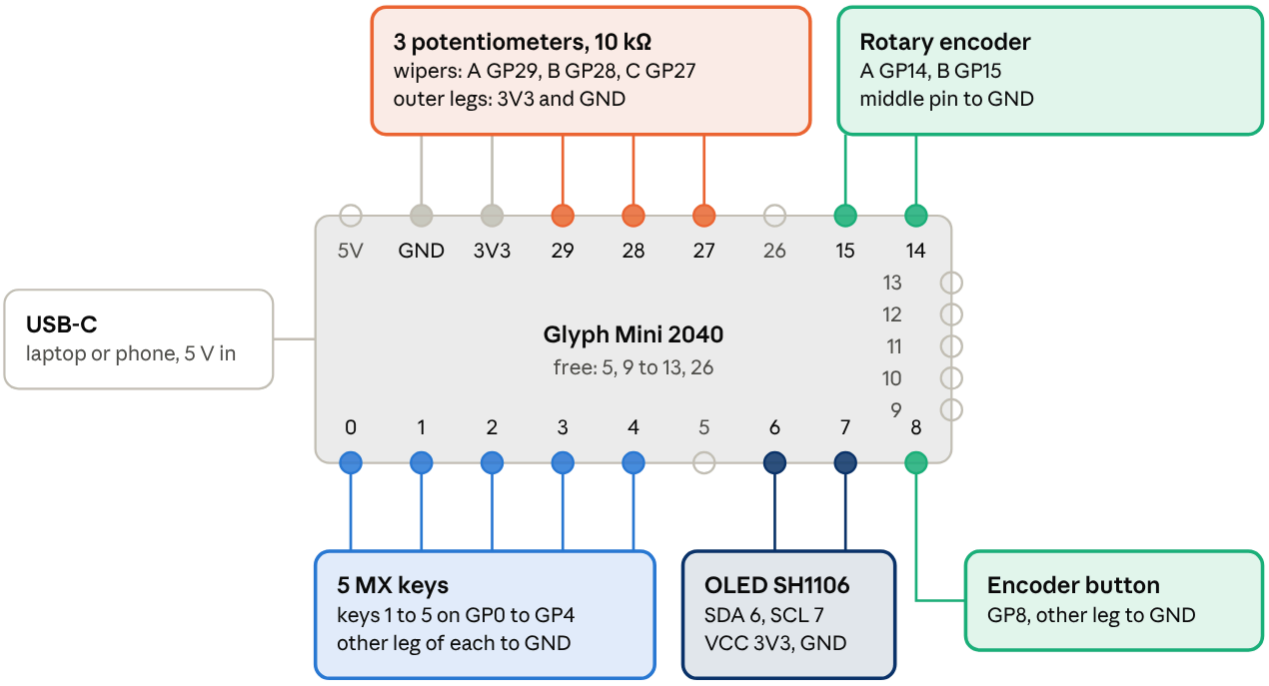
Rules the RP2040 imposes:

- Only GP26–GP29 read analog voltages, so the pots must use them.
- I2C pins come in fixed pairs per controller. GP6 (SDA) + GP7 (SCL) work; GP5 + GP6 do not, because they belong to different I2C controllers.
- Never connect anything to 5V or USB: every input here takes 3.3 V at most.

Wiring drawing

Each part connects to its own pins; the colour of a pad matches the part it serves.

Every part wires straight to one Glyph pin, with shared 3V3 and GND



wiring · pads as laid out on the board, colour per part

Pads are drawn where they sit on the board: 5V to 14 along the top, 13 to 9 down the right, 0 to 8 along the bottom. Hollow pads are unused. Keys, encoder, button, OLED and pots all also need a GND wire; the OLED and pots also need 3V3.

Connections

Every part shares one GND rail; only the OLED and the pots also need the 3V3 rail.

Keys

Key	Leg 1	Leg 2	MIDI note
1	GP0	GND	61 (C#4)
2	GP1	GND	64 (E4)
3	GP2	GND	66 (F#4)
4	GP3	GND	68 (G#4)
5	GP4	GND	73 (C#5)

The OLED shows these notes one octave lower (C#3 for note 61): Adafruit's code counts octaves from C-2.

Rotary encoder

Encoder pin	Glyph Mini 2040
A (outer pin, 3-pin side)	GP14
Common (middle pin, 3-pin side)	GND
B (outer pin, 3-pin side)	GP15
Switch pin 1 (2-pin side)	GP8
Switch pin 2 (2-pin side)	GND

OLED (SH1106 1.3")

OLED pin	Glyph Mini 2040
GND	GND
VCC	3V3
SCL	GP7
SDA	GP6

Potentiometers (10 kΩ)

Knob	Outer leg	Wiper (middle leg)	Outer leg	Sends
Pot A	GND	GP29	3V3	CC 1, modulation
Pot B	GND	GP28	3V3	pitch bend
Pot C	GND	GP27	3V3	CC 11, expression

The code reads the GND end as 127 and the 3V3 end as 0. Swap a pot's two outer legs to reverse it.

Power

The USB-C cable powers everything. The Glyph's 3V3 regulator feeds the OLED and pots; nothing uses 5V.

Firmware

The board runs CircuitPython 10.3.1 with `code.py` on its CIRCUITPY drive; saving a new `code.py` there restarts the board with it, no flashing step.

Files

File	What it is
`KeysetMini2040/code.py`	the keyset code (on the board and in the project folder)
`KeysetMini2040/main.bmp`	the OLED main-screen picture
`KeysetMini2040/original_board_code.py`	the test code that was on the board before
`CIRCUITPY/lib/`	adafruit_midi, adafruit_displayio_sh1106, adafruit_displayio_ssd1306, adafruit_display_text, adafruit_display_shapes, adafruit_imageload, adafruit_debouncer, adafruit_ticks, simpleio

Settings at the top of `code.py`

Setting	Value	Effect
`POTS_CONNECTED`	True	reads the three pots; False silences them
`POT_B_PITCH_BEND`	True	Pot B sends pitch bend instead of a CC
`POT_C_INSTRUMENTS`	False	True makes Pot C pick instruments (Program Change, drums on channel 10)
`POT_DEADBAND`	2	steps a knob must move before it sends; raise to 3 if a knob jitters
`OLED_DRIVER` / `OLED_ADDRESS`	"SH1106" / 0x3C	display chip and I2C address
`OLED_SDA` / `OLED_SCL`	GP6 / GP7	display pins

Setting	Value	Effect
`OLED_CONTRAST`	0xFF	display brightness, 0x01–0xFF

MIDI messages

All messages go out on MIDI channel 1 over USB.

Control	Message	Range
Keys 1–5	Note On / Note Off, velocity 120	notes 61, 64, 66, 68, 73
Pot A (GP29)	Control Change 1, modulation	0–127
Pot B (GP28)	Pitch bend	full range; middle of the pot = no bend
Pot C (GP27)	Control Change 11, expression	0–127
Encoder double-press	Control Change 123, all notes off	—

Each pot reading is the average of 8 samples; values within 2 steps of either end snap to 0 or 127, so a knob left alone stays silent.

Using it

Plug in the USB-C cable and the keyset is ready in about 2 seconds; any app with MIDI input lists it as **Glyph Mini 2040**.

On the keyset

- Press a key to play its note; hold several for a chord.
- Turn the encoder to move the outline between the 3 knob circles and 5 key boxes on the OLED.
- Short-press the encoder to open the selected item's edit screen; short-press again to go back.
- In the edit screen, turn to pick a line, long-press to edit it, turn to change the value, long-press to finish.
- Double-press the encoder to send all notes off.

Edits made with the encoder are lost on restart; defaults live in `code.py`.

Laptop (Ubuntu)

App	Setup	What works
FluidSynth	run `"/home/dinesh/PROJECTS/MIDI CONTROLLER/play.sh"`; it starts FluidSynth, links the keyset and re-links after any drop	keys, modulation, pitch bend, expression
VMPK (on-screen keyboard)	`~/Downloads/vmpk-0.9.2-x86_64.AppImage`; Edit → MIDI Connections: MIDI IN driver ALSA, input Glyph Mini 2040, tick MIDI Thru; output SonivoxEAS or FLUID Synth	shows your key presses; plays through its output
LMMS	Edit → Settings → MIDI: ALSA-Sequencer, restart LMMS; on a track, MIDI input → Glyph Mini 2040	keys, pitch bend; knobs via right-click → Connect to controller
Qsynth	Setup: MIDI driver alsa_seq, audio driver pulseaudio	keys, modulation, pitch bend, expression

To check what the keyset sends, run `aseqdump -p "Glyph Mini 2040"` and press keys or turn knobs.

Phone (Android)

1. Connect with a USB-C to USB-C data cable or an OTG adapter; the phone powers the keyset.
2. In G-Stomper Studio Demo, enable USB MIDI input and pick Glyph Mini 2040.
3. Keys, modulation (Pot A) and pitch bend (Pot B) work straight away.
4. Expression (Pot C) needs MIDI Learn once: long-press the target control, then turn Pot C.

Troubleshooting

Most problems during the build were a wire not reaching its pad; check contact first, code second.

Symptom	Cause	Fix
Nothing works, no MIDI device in the app	board in safe mode: BOOT was held while it started	press RESET without touching BOOT

Symptom	Cause	Fix
MIDI arrives but no sound on the laptop	the keyset is not linked to a synth (links drop when VMPK's MIDI settings change)	run `play.sh`, or `aconnect "Glyph Mini 2040" "FLUID Synth"`
"No OLED at 0x3c … No pull up found"	SDA or SCL wire not touching its pin	reseat or solder the OLED wires on GP6 / GP7
OLED found only with wires swapped	SDA and SCL swapped	SDA → GP6, SCL → GP7; the RP2040 cannot swap them in software
Junk pixels down one edge of the OLED	SH1106 column offset	already fixed in code (`colstart=2`)
Pot reads a steady ~0.5 V or drifts, ignores turning	wiper wire not reaching GP27–GP29	measure at the Glyph pad, not the pot; solder the joint
Pot stuck at exactly 0 V or 3.3 V	wire on an outer leg, not the wiper	move it to the middle leg
Knob floods MIDI while untouched	ADC noise	raise `POT_DEADBAND` to 3
Encoder and its button both dead	encoder common or switch not on GND	check both GND links with a continuity tester
Encoder turns the wrong way	A and B swapped	swap the GP14 and GP15 wires
Upload fails, "port busy"	a serial monitor or another app holds the port	close it, or hold BOOT and tap RESET to get the RPI-RP2 drive
Upload goes to the wrong board	two RP2040 boards plugged in	unplug the one you are not flashing